Teo Kai Jie (Kendrick)

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EDUCATION

Nanyang Technological University **Expected Jul 2028**
Bachelor of Computing (Honours) in Computer Science *Singapore*

- **Current GPA: 4.56/5.00** (Highest Distinction)

Singapore Polytechnic **Mar 2022**
Diploma in Information Technology *Singapore*

- **Final GPA: 3.80/4.00**

EXPERIENCE

Singapore Armed Forces **Dec 2022 – Aug 2024, May 2025 – Jul 2025**
Software Engineer (NSF) *Singapore*

- Developed frontend and backend services for an email-based QR attendance web app using **Node.js**, **Express.js**, and **EJS**, automating check-in workflows for **300+ personnel** in real time, earning a feature on Army News.
- Conducted functional **QA testing** across Python-based Telegram chatbots, diagnosing message-handling defects and isolating edge cases to maintain operational stability.

EcoExchange Pte. Ltd. **Oct 2021 – May 2022**
Lead Frontend Developer (Industry Client Project) *Singapore*

- **Directed a 3-engineer frontend team** using **HTML5**, **CSS3**, and **JavaScript**, managing code reviews and version control in **Git** to deliver a centralized corporate intranet.
- Replaced manual paper-based filing procedures with an interactive ticket-tracking web portal, successfully managing and processing **1,000+ live operational tickets**.

Leaptron Engineering Pte. Ltd. **Feb 2021 – Aug 2021**
Software Engineer (Intern) *Singapore*

- Architected a warehouse management backend using **Node.js** and **SQL**, designing normalized **MySQL** database schemas and stored procedures to handle inventory records and improve query speeds.
- Built a mobile inventory scanner using **Android SDK (Java)**, eliminating manual data entry errors for warehouse staff during inbound shipments.

PROJECTS

Personal Website with AI Chatbot | *Website* | *GitHub* **May 2026 - Aug 2026**

- Engineered a responsive web application using **Next.js**, **TypeScript**, and **Tailwind CSS**, version-controlled with **Git** and deployed on **Vercel**.
- Integrated **RESTful APIs**, connecting the backend to **Gemini API** and **ElevenLabs API** to stream real-time conversational voice responses.
- Implemented an interactive 3D voxel graphics viewer using **Three.js** and **WebGL**, optimizing the rendering pipeline with batch processing to maintain **~60 FPS**.

CellCollab | *Website* | *GitHub* **May 2025 - Oct 2025**

- Engineered a real-time multiplayer web canvas deployed on **Vercel** and **Render** using **React.js** and **Socket.IO**, achieving **sub-100 ms** data synchronization for concurrent user edits and cursor positions.
- Designed a **Node.js** simulation engine with user-configurable tick speeds, calculating grid states and broadcasting synchronized updates to room participants.
- Wrote a **Python** script to parse the Conway Life Lexicon ASCII release and convert pattern assets into 2D binary arrays for brush presets.

SKILLS

- **Languages**: TypeScript, JavaScript, Python, Java, SQL, HTML5, CSS3
- **Frontend:** React.js, Next.js, Tailwind CSS, Three.js, EJS
- **Backend & Databases:** Node.js, Express.js, RESTful APIs, Socket.IO, MySQL
- **Tools & Platforms:** Git, GitHub, Android SDK, Vercel